

$\mathbb{Z}[\pi]$ -module G corresponding to E , by identifying $C^n(X; E)$ with $\text{Hom}_{\mathbb{Z}[\pi]}(C_n(\tilde{X}), G)$ in the following way. An element $\varphi \in C^n(X; E)$ assigns to each $\sigma: \Delta^n \rightarrow X$ a lift to E . Regarding E as the quotient of $\tilde{X} \times G$ under the diagonal action of π , a lift of σ to E is the same as an orbit of a lift to $\tilde{X} \times G$. Such an orbit is a function f assigning to each lift $\tilde{\sigma}: \Delta^n \rightarrow \tilde{X}$ an element $f(\tilde{\sigma}) \in G$ such that $f(\gamma\tilde{\sigma}) = \gamma f(\tilde{\sigma})$ for all $\gamma \in G$, that is, an element of $\text{Hom}_{\mathbb{Z}[\pi]}(C_n(\tilde{X}), G)$.

The basic properties of ordinary cohomology in §3.1 extend without great difficulty to cohomology groups with local coefficients. In order to define the map $f^*: H^n(X'; E') \rightarrow H^n(X; E)$ induced by a bundle map as before, it suffices to observe that a singular simplex $\sigma: \Delta^n \rightarrow X$ and a lift $\tilde{\sigma}': \Delta^n \rightarrow E'$ of $f\sigma$ define a lift $\tilde{\sigma} = (\sigma, \tilde{\sigma}'): \Delta^n \rightarrow f^*(E)$ of σ . To show that $f \simeq g$ implies $f^* = g^*$ requires some modification of the proof of the corresponding result for ordinary cohomology in §3.1, which proceeded by dualizing the proof for homology. In the local coefficient case one constructs a chain homotopy P^* satisfying $g^\# - f^\# = P^*\delta + \delta P^*$ directly from the subdivision of $\Delta^n \times I$ used in the proof of the homology result. Similar remarks apply to proving excision and Mayer-Vietoris sequences for cohomology with local coefficients. To prove the equivalence of simplicial and cellular cohomology with singular cohomology in the local coefficient context, one should use the telescope argument from the proof of Lemma 2.34 to show that $H^n(X^k; E) \approx H^n(X; E)$ for $k > n$. Once again details will be left to the reader.

The difference between homology with local coefficients and cohomology with local coefficients is illuminated by comparing the following proposition with our earlier identification of $H_*(X; \mathbb{Z}[\pi_1 X])$ with the ordinary homology of the universal cover of X .

Proposition 3H.5. *If X is a finite CW complex with universal cover $\tilde{X}$ and fundamental group π , then for all n , $H^n(X; \mathbb{Z}[\pi])$ is isomorphic to $H_c^n(\tilde{X}; \mathbb{Z})$, cohomology of $\tilde{X}$ with compact supports and ordinary integer coefficients.*

For example, consider the n -dimensional torus T^n , the product of n circles, with fundamental group $\pi = \mathbb{Z}^n$ and universal cover $\mathbb{R}^n$. We have $H_i(T^n; \mathbb{Z}[\pi]) \approx H_i(\mathbb{R}^n)$, which is zero except for a $\mathbb{Z}$ in dimension 0, but $H^i(T^n; \mathbb{Z}[\pi]) \approx H_c^i(\mathbb{R}^n)$ vanishes except for a $\mathbb{Z}$ in dimension n , as we saw in Example 3.34.

To prove the proposition we shall use a few general facts about cohomology with compact supports. One significant difference between ordinary cohomology and cohomology with compact supports is in induced maps. A map $f: X \rightarrow Y$ induces $f^\#: C_c^n(Y; G) \rightarrow C_c^n(X; G)$ and hence $f^*: H_c^n(Y; G) \rightarrow H_c^n(X; G)$ provided that f is **proper**: The preimage $f^{-1}(K)$ of each compact set K in Y is compact in X . Thus if $\varphi \in C_c^n(Y; G)$ vanishes on chains in $Y - K$ then $f^\#(\varphi) \in C_c^n(X; G)$ vanishes on chains in $X - f^{-1}(K)$. Further, to guarantee that $f \simeq g$ implies $f^* = g^*$ we should restrict attention to homotopies that are proper as maps $X \times I \rightarrow Y$. Relative groups

$H_c^n(X, A; G)$ are defined when A is a closed subset of X , which guarantees that the inclusion $A \hookrightarrow X$ is a proper map. With these constraints the basic theory of §3.1 translates without difficulty to cohomology with compact supports.

In particular, for a locally compact CW complex X one can compute $H_c^*(X; G)$ using *finite cellular cochains*, the cellular cochains vanishing on all but finitely many cells. Namely, to compute $H_c^n(X^n, X^{n-1}; G)$ using excision one first has to identify this group with $H_c^n(X^n, N(X^{n-1}); G)$ where $N(X^{n-1})$ is a closed neighborhood of X^{n-1} in X^n obtained by deleting an open n -disk from the interior of each n -cell. If X is locally compact, the obvious deformation retraction of $N(X^{n-1})$ onto X^{n-1} is a proper homotopy equivalence. Hence via long exact sequences and the five-lemma we obtain isomorphisms $H_c^n(X^n, X^{n-1}; G) \approx H_c^n(X^n, N(X^{n-1}); G)$, and by excision the latter group can be identified with the finite cochains.

Proof of 3H.5: As noted above, we can compute $H_c^*(\tilde{X}; \mathbb{Z})$ using the groups $C_f^n(\tilde{X}; \mathbb{Z})$ of finite cellular cochains $\varphi: C_n \rightarrow \mathbb{Z}$, where $C_n = H_n(\tilde{X}^n, \tilde{X}^{n-1})$. Giving $\tilde{X}$ the CW structure lifting the CW structure on X , then since X is compact, finite cellular cochains are exactly homomorphisms $\varphi: C_n \rightarrow \mathbb{Z}$ such that for each cell e^n of $\tilde{X}$, $\varphi(\gamma e^n)$ is nonzero for only finitely many covering transformations $\gamma \in \pi$. Such a φ determines a map $\hat{\varphi}: C_n \rightarrow \mathbb{Z}[\pi]$ by setting $\hat{\varphi}(e^n) = \sum_{\gamma} \varphi(\gamma^{-1} e^n) \gamma$. The map $\hat{\varphi}$ is a $\mathbb{Z}[\pi]$ -homomorphism since if we replace the summation index γ in the right side of $\varphi(\eta e^n) = \sum_{\gamma} \varphi(\gamma^{-1} \eta e^n) \gamma$ by $\eta \gamma$, we get $\sum_{\gamma} \varphi(\gamma^{-1} e^n) \eta \gamma$. The function $\varphi \mapsto \hat{\varphi}$ defines a homomorphism $C_f^n(\tilde{X}; \mathbb{Z}) \rightarrow \text{Hom}_{\mathbb{Z}[\pi]}(C_n, \mathbb{Z}[\pi])$ which is injective since φ is recoverable from $\hat{\varphi}$ as the coefficient of $\gamma = 1$. Furthermore, this homomorphism is surjective since a $\mathbb{Z}[\pi]$ -homomorphism $\psi: M \rightarrow \mathbb{Z}[\pi]$ has the form $\psi(x) = \sum_{\gamma} \psi_{\gamma}(x) \gamma$ with $\psi_{\gamma} \in \text{Hom}_{\mathbb{Z}}(M, \mathbb{Z})$ satisfying $\psi_{\gamma}(x) = \psi_1(\gamma^{-1} x)$, so ψ_1 determines ψ . The isomorphisms $C_f^n(\tilde{X}; \mathbb{Z}) \approx \text{Hom}_{\mathbb{Z}[\pi]}(C_n, \mathbb{Z}[\pi])$ are isomorphisms of cochain complexes, so the respective cohomology groups $H_c^n(\tilde{X}; \mathbb{Z})$ and $H^n(X; \mathbb{Z}[\pi])$ are isomorphic. $\square$

Cup and cap product work easily with local coefficients in a bundle of rings, the latter concept being defined in the obvious way. The cap product can be used to give a version of Poincaré duality for a closed n -manifold M using coefficients in a bundle of rings E under the same assumption as with ordinary coefficients that there exists a fundamental class $[M] \in H_n(M; E)$ restricting to a generator of $H_n(M, M - \{x\}; E)$ for all $x \in M$. By excision the latter group is isomorphic to the fiber ring R of E . The same proof as for ordinary coefficients then shows that $[M] \frown: H^k(M; E) \rightarrow H_{n-k}(M; E)$ is an isomorphism for all k .

Taking R to be one of the standard rings $\mathbb{Z}$, $\mathbb{Q}$, or $\mathbb{Z}_p$ does not give anything new since the only ring automorphism these rings have is the identity, so the bundle of rings E must be the product $M \times R$. To get something more interesting, suppose we take R to be the ring $\mathbb{Z}[i]$ of Gaussian integers, the complex numbers $a + bi$ with

$a, b \in \mathbb{Z}$. This has complex conjugation $a + bi \mapsto a - bi$ as a ring isomorphism. If M is nonorientable and connected we can use the homomorphism $\omega : \pi_1(M) \rightarrow \{\pm 1\}$ that defines the bundle of groups $M_{\mathbb{Z}}$ to build a bundle of rings E corresponding to the action of $\pi_1(M)$ on $\mathbb{Z}[i]$ given by $\gamma(a + bi) = a + \omega(\gamma)bi$. The homology and cohomology groups of M with coefficients in E depend only on the additive structure of $\mathbb{Z}[i]$ so they split as the direct sum of their real and imaginary parts, which are just the homology or cohomology groups with ordinary coefficients $\mathbb{Z}$ and twisted coefficients $\tilde{\mathbb{Z}}$, respectively. The fundamental class in $H_n(M; \tilde{\mathbb{Z}})$ constructed in Example 3H.3 can be viewed as a pure imaginary fundamental class $[M] \in H_n(M; E)$. Since cap product with $[M]$ interchanges real and imaginary parts, we obtain:

Theorem 3H.6. *If M is a nonorientable closed connected n -manifold then cap product with the pure imaginary fundamental class $[M]$ gives isomorphisms $H^k(M; \mathbb{Z}) \approx H_{n-k}(M; \tilde{\mathbb{Z}})$ and $H^k(M; \tilde{\mathbb{Z}}) \approx H_{n-k}(M; \mathbb{Z})$. $\square$*

More generally this holds with $\mathbb{Z}$ replaced by other rings such as $\mathbb{Q}$ or $\mathbb{Z}_p$. There is also a version for noncompact manifolds using cohomology with compact supports.

Exercises

1. Compute $H_*(S^1; E)$ and $H^*(S^1; E)$ for $E \rightarrow S^1$ the nontrivial bundle with fiber $\mathbb{Z}$.
2. Compute the homology groups with local coefficients $H_n(M; M_{\mathbb{Z}})$ for a closed nonorientable surface M .
3. Let $\mathcal{B}(X; G)$ be the set of isomorphism classes of bundles of groups $E \rightarrow X$ with fiber G , and let $E_0 \rightarrow B\text{Aut}(G)$ be the bundle corresponding to the 'identity' action $\rho : \text{Aut}(G) \rightarrow \text{Aut}(G)$. Show that the map $[X, B\text{Aut}(G)] \rightarrow \mathcal{B}(X, G)$, $[f] \mapsto f^*(E_0)$, is a bijection if X is a CW complex, where $[X, Y]$ denotes the set of homotopy classes of maps $X \rightarrow Y$.
4. Show that if finite connected CW complexes X and Y are homotopy equivalent, then their universal covers $\tilde{X}$ and $\tilde{Y}$ are proper homotopy equivalent.
5. If X is a finite nonsimply-connected graph, show that $H^n(X; \mathbb{Z}[\pi_1 X])$ is zero unless $n = 1$, when it is the direct sum of a countably infinite number of $\mathbb{Z}$'s. [Use Proposition 3H.5 and compute $H_c^n(\tilde{X})$ as $\varinjlim H^n(\tilde{X}, \tilde{X} - T_i)$ for a suitable sequence of finite subtrees $T_1 \subset T_2 \subset \cdots$ of $\tilde{X}$ with $\bigcup_i T_i = \tilde{X}$.]
6. Show that homology groups $H_n^{\ell f}(X; G)$ can be defined using **locally finite** chains, which are formal sums $\sum_{\sigma} g_{\sigma} \sigma$ of singular simplices $\sigma : \Delta^n \rightarrow X$ with coefficients $g_{\sigma} \in G$, such that each $x \in X$ has a neighborhood meeting the images of only finitely many σ 's with $g_{\sigma} \neq 0$. Develop this homology theory far enough to show that for a locally compact CW complex X , $H_n^{\ell f}(X; G)$ can be computed using infinite cellular chains $\sum_{\alpha} g_{\alpha} e_{\alpha}^n$.